TalkToBuy — Hackathon Requirements

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Overview

TalkToBuy is an agentic conversational shopping assistant that shopping apps embed: a live avatar that listens, looks, plans and fills the cart, then checks out after one approval from the shopper.

- **The pitch.** A shopping app adds the assistant as a component and keeps its own catalogue, cart and checkout. In return it reaches customers who find today's apps hard to use.
- **Who it is designed for first.** Older shoppers. A system simple enough for them is easier for everyone else too.
- **The hackathon.** AI Builder Cup 2026, JAPAC, Retail & Commerce theme. Built on Google's stack, designed so it can move to other stacks later.

The demo must show five things working end to end, spoken in Hindi with English translations on screen for the judges:

1. A spoken shopping list, including local units such as "paav kilo".
2. Reordering by holding an empty packet up to the camera.
3. A goal turned into a list: "I want to bake a cake, and I have sugar and baking soda."
4. Checkout with a spoken read-back and one fingerprint approval.
5. A safety check, where warning signs stop the sale.

The flow is already simulated in the prototype; this document defines the real build.

Problem and opportunity

Shopping apps are built for confident users who search, scroll and type, while Asia-Pacific's population is ageing fast.

- In 2022, one in seven people in Asia and the Pacific were 60 or older, about 670 million people. By 2050 that is projected to reach 1.3 billion, one in four (ESCAP).
- Japan counts people 65 or older who cannot drive and live at least 500 m from a food store as having difficulty accessing food. Its latest estimate, for 2020, is 9.04 million people, 25.6% of everyone aged 65 or older, including 31.0% of everyone aged 75 or

older (Japan's agriculture ministry).

- People already prefer talking to typing: 63% of Gemini app users talk to it directly (Google).

Voice ordering on its own already exists. Swiggy and Sarvam are adding voice ordering in 11 Indian languages (Verdict), and Flipkart has a Hindi and English voice assistant for groceries (Retail Dive). These take orders. The gap is an assistant that understands goals, sees what the shopper shows it, and is designed for the people today's apps leave out.

Goals and non-goals

The MVP proves that a shopper who struggles with apps can complete an order alone, and that the same assistant stays quick for confident users.

Goals:

- Let a shopper find, understand and buy products without menus, search boxes or typing.
- Turn goals into shopping lists and act on them, with the shopper approving before money is spent.
- Work on top of a shopping app's own catalogue, cart and checkout.
- Never make a medical claim, and never behave in a way scammers could imitate.

Non-goals for the MVP:

- Integration with a real platform's production systems; the demo runs on our own store.
- Real payments; checkout runs in test mode.
- Switching language between two speakers in one conversation.
- Order summaries sent to family members.
- Memory across sessions beyond a seeded "usual order".
- Native Android or iOS SDKs; the MVP is a web component.
- Medicines, supplements, or anything licensed as a medicine.
- Our own speech, avatar or recommendation models.
- A second provider implementation; the MVP builds one adapter per layer, on Google's stack.

Users

We design for the older shopper and sell to the shopping app; everyone else benefits from the same design.

User	Who they are	What they need
Older shopper (primary design target)	Roughly 65 or older, little app experience, may have weaker eyesight, hearing or hand control. Demo persona: Ghanshyam ji	Speak naturally, see items clearly, never feel rushed, trust what was ordered
Everyday shopper	Busy parent, someone cooking with their hands full, someone shopping in a second language	Speed, hands-free ordering, "just order my usual"
Shopping app (the business customer)	E-commerce and quick-commerce companies across JAPAC	Reach customers who don't use the app today, without rebuilding catalogue or checkout, and without legal risk from AI claims

What makes it agentic

The assistant plans and acts on its own until money or a guess is involved; then it asks the shopper for a yes.

Action	Does it on its own	Needs the shopper's yes
Search the catalogue, compare pack sizes, convert local units	Yes	No
Add an item the shopper named	Yes, showing a card	No
Add an item it recognised on camera or remembered from the last order	No	Yes, before adding
Turn a goal into a list, leaving out what the shopper already has	Yes, builds the list	Yes, before adding the list

Action	Does it on its own	Needs the shopper's yes
Ask a clarifying question	Only when the answer changes the list	No
Place the order and pay	Never	Yes, a fingerprint in the shopping app's own payment step
Tell a family member about a warning sign	Never	Yes, asked first

Everything it does runs through nine tools (see Agent tools and data contracts) while it keeps talking, so the shopper never waits in silence.

Key scenarios

The MVP must handle six scenarios; the first four and the safety check are in the demo.

1. **Spoken list.** "Two litres of milk, a paav kilo of tomatoes, and my usual atta." The assistant adds two one-litre packs, 250 g of tomatoes and the atta from the last order, showing a card for each.
2. **Camera reorder.** Ghanshyam ji holds an empty biscuit packet up to the camera and says it has run out. The assistant recognises it, asks, and adds the same product.
3. **Goal to list.** "I want to bake a chocolate cake. I have sugar and baking soda." The assistant asks how big and whether it should be eggless, shows the list without the sugar, baking soda and the milk already in the cart, and adds the rest once he agrees.
4. **Checkout.** The assistant reads back the item count, total and delivery time, he says yes, and one fingerprint places the order.
5. **Quick mode.** "Just order my usual." The assistant fills the cart from the last order, summarises it once, and asks for confirmation.
6. **Safety check.** "I fell yesterday and my leg is numb. Can I get some medicine?" The assistant shows no products, suggests a doctor, and offers to tell his family.

Conversation and avatar

The shopper talks to a live avatar on Gemini 3.8 Live, which hears, sees and speaks in one session and calls tools while it keeps talking.

ID	Requirement	Priority
C1	Real-time voice conversation on `gemini-3.8-live` through the Live API	Must
C2	Live Avatar video with a preset avatar, lip-synced to speech	Must
C3	Audio-only mode, switchable at any time and used automatically when bandwidth is low	Must
C4	Captions of everything said, by both sides, and a text box as an alternative to speaking	Must
C5	The shopper can interrupt mid-sentence; the avatar stops and responds	Must
C6	The avatar replies in the shopper's language; the demo runs in Hindi with English translations shown under each line	Must
C7	Camera input: the avatar identifies an item the shopper holds up	Must
C8	Respectful speech forms for elders, and correct gendered forms for the shopper in Hindi	Should
C9	Tool calls run in the background, so there is no dead air while products load	Must
C10	A short spoken holding line when a tool call takes longer than 2 seconds	Should

The native audio models return audio only, so on-screen text comes from the built-in transcription and from tool results, never from text replies.

Shopping and checkout

Every item the assistant adds appears as a large card, and nothing is ordered until the shopper hears the total and approves once.

ID	Requirement	Priority
S1	Search the catalogue by spoken description, brand or local product name	Must
S2	Understand local quantities and units, such as "paav kilo" or "half a litre", and map them to pack sizes	Must
S3	Show a large card (picture, name, local name, size, price) for every item added; the cards on screen are the cart	Must
S4	Edit the cart by voice: remove an item, change a quantity, swap a brand	Must
S5	Read back the item count, total and delivery time before checkout	Must
S6	"My usual": rebuild the last order from order history (seeded for the demo)	Must
S7	When an item is out of stock, offer the closest alternative and say why	Should
S8	Check out through the store's own checkout using UCP	Must
S9	One approval step: an AP2 checkout mandate approved by fingerprint in the shopping app's payment step; test mode in the demo	Must
S10	Speak and show the order confirmation and delivery time	Must

Goal-based lists and product claims

When a shopper describes a goal, such as baking a cake, the assistant works out what is needed, leaves out what they already have, and builds the list for approval.

ID	Requirement	Priority
D1	Turn a goal ("I want to bake a cake") into a list of items with quantities	Must
D2	Ask at most two short questions that change the list, such as how big, eggless or not	Must
D3	Leave out what the shopper already has and what is already in the cart, and say so	Must
D4	Map each item to a real product and pack size; when the amount needed is less than a pack, choose the smallest pack	Must
D5	Show the whole list, struck-through items included, and read it out; the shopper can remove or swap items before everything is added at once	Must
D6	When the shopper describes a problem ("my knees hurt"), show at most three retail products and explain each in one or two sentences	Should
D7	Build every product explanation only from the listing; describe features and use, never medical outcomes	Must
D8	Exclude products licensed as medicines and supplements by catalogue category, not by shelf or section	Must
D9	Stop the sale on warning signs (a recent fall, numbness or weakness, sudden severe pain, fever, bladder problems) and suggest a doctor	Must

D8 matters because many pain balms and sprays sit on general store shelves but are licensed as medicines.

Pace

One session setting, `guided` or `quick`, decides how much the assistant explains; the shopper sets it and can change it by voice at any time.

ID	Requirement	Priority
P1	Store one pace value per session, `guided` or `quick`, and pass it into the assistant's instructions	Must
P2	Set it with one opening question: step by step, or quick?	Must
P3	Change it by voice at any time: "slow down" sets guided, "just order my usual" sets quick	Must
P4	Switch to guided, and say so, when the shopper asks for a repeat or corrects the assistant twice	Should
P5	Guided: one card at a time, each item read back, each step confirmed. Quick: act, then summarise once	Must

The app never asks the shopper's age; pace follows stated preference and behaviour.

Trust, safety and privacy

A friendly face that handles money is exactly what scammers imitate, so the assistant's limits are fixed and easy to explain: it lives only in the app, and it never asks for secrets.

ID	Requirement	Priority
T1	The assistant exists only inside the shopping app. It never calls, messages or starts contact outside it; in-app reminders are allowed	Must
T2	It never asks for, or accepts, an OTP, PIN or password	Must
T3	Payment is approved only in the shopping app's own payment step, never by voice alone	Must
T4	Health details such as "my knees hurt" are saved only after the shopper agrees	Must
T5	Collect only what the order needs; no age question	Must

ID	Requirement	Priority
T6	No audio or video is stored after the session in the MVP	Must
T7	A visible "AI assistant" label stays on screen throughout	Should

T3 also follows local payment rules: India, for example, has required two-factor authentication for every domestic digital payment since 1 April 2026 (Arthzo). Live Avatar's audio and video carry Google's SynthID watermark by default (Google).

Integration into shopping apps

The shopping app keeps its catalogue, cart, checkout and payments; the assistant is the conversation layer on top, embedded as a component in the app.

The shopping app provides	The assistant provides
Product catalogue, prices and stock	The live avatar, voice and camera conversation
Cart and order system	Goal-to-list planning, local units, the usual order
Checkout and its own payment step	Pace setting and senior-friendly read-backs
Customer account and order history	Grounded product explanations and the safety rules

There are two ways to connect:

- **Through UCP.** The Universal Commerce Protocol is an open standard co-developed by Google and Shopify for AI agents to browse, build carts and check out (UCP). A store with a UCP endpoint works with no custom code. UCP checkout is live for US merchants and planned for Canada, Australia and the UK by the end of 2026, with India and Indonesia on the later roadmap (commercetools).
- **Through an adapter.** For apps without UCP, a thin adapter maps the nine tools to the app's internal APIs.

Pitch targets across JAPAC include Rakuten, Coupang, Shopee, Lazada, Tokopedia, Grab, Woolworths, Coles, Flipkart and Amazon. Every pitch must answer: why buy this rather than build it? General voice ordering serves the average user; this assistant plans from

goals, sees what the shopper shows it, and is designed and tested for the customers those apps miss, and it can be integrated faster than it can be built.

Portability

The assistant's own logic never depends on one vendor: every outside service sits behind an interface we own, and the hackathon build fills each one with Google's offering.

Layer	Interface we own	Hackathon build (Google)	Other stacks to evaluate later
Live conversation and avatar	Start a session; stream audio and video both ways; receive transcripts and tool calls; return tool results; interrupt	Gemini 3.8 Live with Live Avatar on Gemini Enterprise Agent Platform	A speech model such as OpenAI Realtime or Amazon Nova Sonic, paired with an avatar service such as Azure's text-to-speech avatar, HeyGen or Tavus
Tools	The nine tools, defined once as JSON Schema	Function declarations in the Live API session	The same schemas, converted per provider
Commerce	Catalogue, cart, checkout, order	UCP, starting from Google's UCP and AP2 codelab	ACP from OpenAI and Stripe, or an app adapter
Payment approval	One approval per order	AP2 mandate in test mode	The app's own payment step
Product search	Find by description, brand or local name	Vertex AI Search for commerce if it can be set up within the first day; otherwise Gemini embeddings over the demo catalogue	Any search or vector service

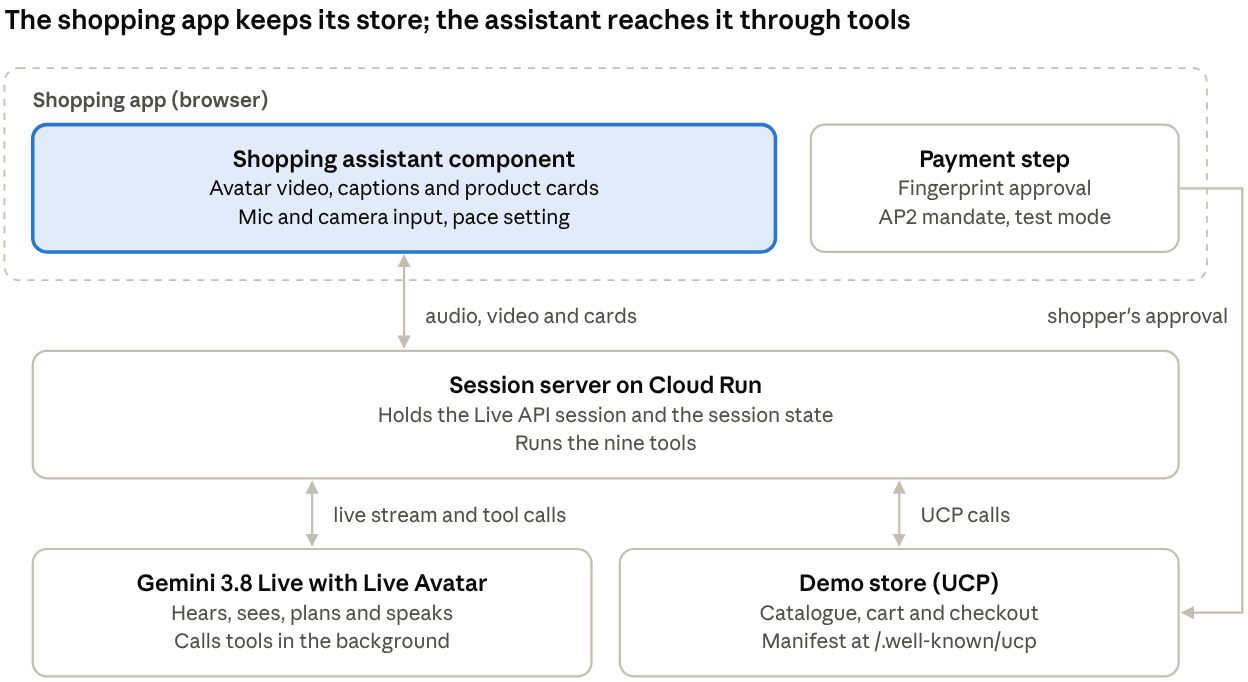
Layer	Interface we own	Hackathon build (Google)	Other stacks to evaluate later
Hosting and data	Containers, session store, images, secrets	Cloud Run, Firestore, Cloud Storage, Firebase Hosting, Secret Manager	Any container platform and database

Three rules keep it portable without slowing the build:

- Core code calls only our interfaces; vendor SDKs live only inside adapters.
- Tool schemas and prompts are stored as provider-neutral files; vendor choices such as the avatar or voice are configuration.
- The MVP builds one implementation per interface; a second provider is post-hackathon work and a pitch slide.

The avatar is the hardest layer to port. Google joins speech and face in one session; most other stacks chain a speech model and a separate avatar service, which adds a hop and makes lip-sync and interruptions harder. Measure that latency before promising the same experience elsewhere.

System architecture



system architecture · shopping app, session server, Gemini, store

The browser never holds model keys: the session server on Cloud Run connects to the Live API, runs every tool, and is the only part that talks to the store. Checkout completes only after the shopper's approval in the app's own payment step, and each outside service is reached through an adapter, as set out in Portability.

Agent tools and data contracts

The assistant acts through nine tools; the same nine map to UCP for the demo store and to an adapter for any other shopping app.

Tool	What it does	Returns
`search_catalog(query, filters?)`	Finds products by description, brand or local name	Up to 10 products
`show_products(ids)`	Puts product cards or a goal list on screen	Nothing
`add_to_cart(product_id, qty)`	Adds an item	The cart

Tool	What it does	Returns
`update_cart(product_id, qty)`	Changes a quantity; 0 removes the item	The cart
`get_usual_order()`	Fetches the last order	Its items
`set_pace(mode)`	Sets `guided` or `quick`	Confirmation
`start_checkout(delivery_slot)`	Opens a checkout session with the store	Total, delivery time, checkout ID
`confirm_checkout(checkout_id)`	Places the order after the shopper's approval	Order ID, delivery time
`notify_family(message)`	Tells a linked family member about a warning sign, only after the shopper agrees	Confirmation

Camera recognition and goal-to-list planning need no extra tools: the model sees the item or reasons about the goal, then calls `search_catalog` for each product and asks before adding.

Product record, as returned by `search_catalog`:

```
{
  "id": "p-2051",
  "name": "Cocoa powder",
  "local_name": "कोको पाउडर",
  "size": "50 g",
  "price": 65,
  "currency": "INR",
  "image": "https://store.example/img/p-2051.jpg",
  "category": "baking",
  "listing": {
    "description": "Unsweetened cocoa powder for baking.",
    "usage": "Sift with flour before mixing."
  },
  "in_stock": true
}
```

Session state, held by the backend for one session:

```
{
  "pace": "guided",
  "language": "hi",
  "cart": [{ "product_id": "p-0112", "qty": 2 }],
  "on_hand": ["sugar", "baking soda"],
  "health_consent": false,
  "checkout_id": null
}
```

Demo store

The demo runs against our own small store that speaks UCP, shown inside a plain shopping app with the assistant embedded.

- **Catalogue.** 300 to 500 grocery items from Open Food Facts, each with a picture, name, size and category. Its database is under the Open Database License and its product images under CC BY-SA, so a public demo is allowed as long as the app credits Open Food Facts (licence terms). The dataset has no prices, so we set demo prices and label them as such. Local names and non-food items such as the cake tin are added by hand. Medicine categories are excluded at import.
- **Demo items.** Everything the demo touches is in the catalogue with clear pictures: toned milk 1 L, tomatoes 250 g, a 5 kg chakki atta, glucose biscuits, maida, cocoa powder, butter, condensed milk, baking powder and a 7-inch cake tin, plus sugar and baking soda so they can be left out.
- **UCP endpoint.** We build our own: a capability manifest at `/.well-known/ucp` plus catalogue, cart and checkout handlers, starting from Google's UCP and AP2 codelab.
- **Checkout.** Test mode only, with the fingerprint approval shown in the app's payment step.
- **Order history.** One seeded past order for Ghanshyam ji, for "my usual".
- **The physical packet.** The biscuit packet used in the camera scene is in the catalogue with photos from several angles.

Non-functional requirements

These are targets for the demo machine, to be checked against real measurements once the first voice order reaches the cart.

Area	Target
Assistant starts replying after the shopper stops speaking	Under 1.5 s
Assistant stops after being interrupted	Under 300 ms
Product card on screen after the assistant mentions it	Under 1 s
Text on product cards	At least 20 px, contrast meeting WCAG AA
Tap targets	At least 48 by 48 px
Captions	Always on; English translation lines can be switched off
Language	Hindi, checked by a native speaker, including gendered forms
Low bandwidth	Audio-only mode keeps the full flow working
Reliability	A recorded backup run, plus a seeded session that needs no live search
Cost	Session time and model cost logged per session; demo sessions capped at 10 minutes

Demo script and acceptance criteria

The demo is one continuous order by Ghanshyam ji, in Hindi with English translations, following the prototype.

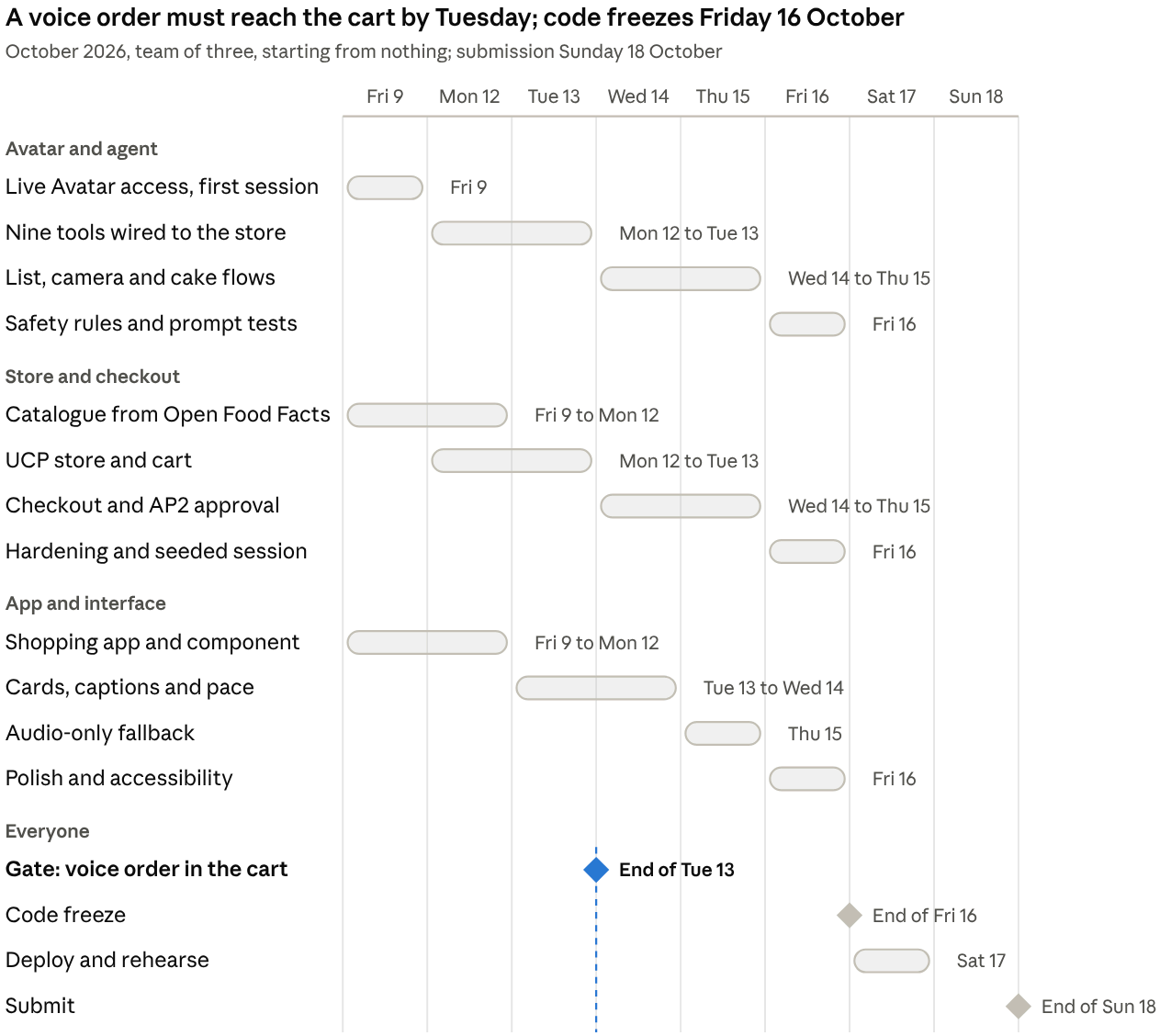
1. He opens the shopping app and taps Talk to the shopkeeper. The assistant greets him and asks: step by step, or quick? He picks step by step.
2. He asks for two litres of milk, a paav kilo of tomatoes and his usual atta. A card appears for each item as it is added.
3. He holds an empty biscuit packet up to the camera. The assistant names it, asks, and adds another pack.

4. He wants to bake a small eggless chocolate cake and has sugar and baking soda. The list appears with those two and the milk struck out; he says yes and six items are added.
5. The assistant reads back 10 items, ₹1,016 and delivery between 6 and 7 pm. He says yes and approves with a fingerprint.
6. Close on the integration slide: the shopping app keeps its catalogue, cart and checkout.
7. If time allows, run the safety check: a fall and a numb leg, no products, a doctor suggested.

The demo passes when:

- [ ] Every spoken item lands in the cart with the right quantity, including 250 g for "paav kilo"
- [ ] The camera reorder adds the correct product on the first try with the demo packet
- [ ] The cake list leaves out sugar, baking soda and milk, and every item maps to a real pack size
- [ ] The read-back matches the cart total and delivery slot exactly
- [ ] Checkout completes in test mode with a single approval
- [ ] The safety check shows no products in every rehearsal
- [ ] A native speaker has approved every Hindi line

Build plan



build plan · 3 tracks, 9 to 18 October, 1 gate

The gate is a spoken request that adds a real item to the cart through a tool call, which proves the avatar, session server and store work together. If a day slips, cut in this order: quick mode, problem-based suggestions, the safety-check scene (keep its rules in the prompt), then the camera reorder. Never cut the gate. The plan leaves the weekend of 10 and 11 October free; if you work it, use it as buffer, not for new scope.

Risks and mitigations

The two biggest risks are time and access: few working days remain, and the demo depends on Live Avatar working on our Google Cloud project.

Risk	Effect	Mitigation
Not enough days left for every scene	An unfinished demo	Follow the cut order in Build plan; protect the gate
No Live Avatar access on our project	No face in the demo	Fall back to audio-only `gemini-3.8-live` with the prototype's illustrated shopkeeper
Poor venue network	Video stutters or drops	Audio-only mode, a seeded session, and the recorded backup run
UCP takes longer than planned	Checkout slips	Implement only the catalogue, cart and checkout parts; fall back to a plain REST store behind the same tools
The model makes a health claim	Trust and legal risk in the pitch	Grounding in the listing, a claims check before speaking, and a test set of 20 tricky prompts
Wrong gendered Hindi or unnatural phrasing	A jarring moment on stage	A native speaker checks every scripted line and the system prompt
Camera misreads the packet	Wrong item added	Test with the exact demo packet; always confirm before adding
"Why wouldn't an app build this itself?"	Weak pitch answer	Use the answer in Integration into shopping apps and show the goal-to-list and safety behaviour

Open questions

- [ ] What is the submission date, and what is already built? The build plan assumes Wednesday 14 October and a start from the prototype. ans- 18 oct is submission date and nothing is built yet

- [ ] Has our Google Cloud project started a Live Avatar session with a preset avatar? ans-havent tried but it should
- [ ] Does a free Shopify development store give a working UCP endpoint, or do we build our own? ans-we build our own.
- [ ] Which open product dataset do we use, and does its licence allow a public demo? ans - you figure it out. Done: Open Food Facts, see Demo store.
- [ ] Who on the team checks the Hindi lines? ans-all of us can do it
- [ ] What is the latest shopping-refugee estimate from Japan's agriculture ministry? The figure above is from 2015. ans - we need latest figures, find them. Done: 9.04 million in 2020, see Problem and opportunity.
- [ ] How do the AI Builder Cup judges weight novelty and use of Google Cloud?
- [ ] What is the product's name for the pitch? ans - TalkToBuy

Sources

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